AMBATI MOKESH REDDY

Microsoft Azure & AWS • Networking & Security • Monitoring

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PROFESSIONAL SUMMARY

Recent BCA graduate (2025, CGPA 8.5) seeking an entry-level Cloud Engineer / Site Reliability Engineer role, with hands-on experience across 4 self-driven Azure infrastructure projects covering networking, high-availability deployment, and monitoring.

Skilled in designing secure network architectures (Hub-Spoke topology, 3-tier deployments, VNet Peering, NSGs) and building centralized observability with Azure Monitor and Grafana.

Comfortable across the deployment lifecycle from Infrastructure-as-Code (Terraform) to CI/CD (GitHub Actions); currently pursuing AWS Cloud Practitioner certification to build multicloud capability.

TECHNICAL SKILLS

- **Cloud Platforms:** Microsoft Azure (Networking, Monitoring, Deployment); AWS (Foundational — Cloud Practitioner in progress)
- **AWS (Learning):** EC2, S3, IAM, VPC — hands-on familiarity as part of Cloud Practitioner preparation
- **Networking:** VNet, VPC, Subnetting, NSG, Route Tables, Load Balancers, Application Gateway, Traffic Manager
- **DevOps & IaC:** Git, GitHub, GitHub Actions, Docker, Kubernetes, Terraform
- **Monitoring:** Azure Monitor, Log Analytics Workspace, Azure Managed Grafana, CloudWatch
- **Scripting & OS:** Python (Basic), Bash, PowerShell, Windows, Linux (Ubuntu)
- **Databases:** MySQL, Azure SQL
- **Web Fundamentals:** HTML, CSS

CERTIFICATIONS (4)

- AWS Certified Cloud Practitioner — In Progress
- Introduction to Cloud Computing — IBM (Coursera)
- Cloud Virtualization, Containers and APIs — Duke University (Coursera)
- Introduction to Networking and Storage — IBM (Coursera)

INTERNSHIP EXPERIENCE

Web Development Intern — Bharat Intern

Oct 2024 – Nov 2024

- Designed and implemented responsive front-end interfaces for web applications using HTML, CSS, and JavaScript.
- Delivered functional, cross-browser-consistent pages within a structured development environment.

PROJECT EXPERIENCE (4)

Azure Hub-Spoke Network Architecture with Security, Grafana Monitoring & Smart Alerts

- Designed and implemented an enterprise-grade Hub-Spoke network architecture using Virtual Networks, VNet Peering, NSGs, and Route Tables to enable secure, scalable connectivity across environments.
- Configured a centralized hub VNet for shared services with peered spoke VNets to isolate workloads while maintaining controlled inter-VNet communication.
- Applied NSG rules and User-Defined Routes to enforce least-privilege network segmentation and restrict unnecessary east-west traffic between spokes.
- Integrated Azure Monitor, Log Analytics Workspace, and Azure Managed Grafana to build centralized dashboards providing visibility into network and resource health.
- Configured proactive smart alerts on key metrics and logs to enable faster detection and response to infrastructure issues.

Technologies: *Microsoft Azure, VNet, VNet Peering, NSG, Route Tables, Azure Monitor, Log Analytics, Azure Managed Grafana*

Multi-Cloud 3-Tier Architecture: High-Availability Cloud Infrastructure (Azure & AWS)

- Designed and deployed secure, highly available **3-tier architectures (Web, Application, and Database tiers)** on **Microsoft Azure** and **Amazon Web Services (AWS)** using industry best practices for scalability, reliability, and fault tolerance.
- Configured **Azure VNet** and **AWS VPC** with dedicated subnets, implementing **Azure NSGs**, **AWS Security Groups**, and **Network ACLs** to enable secure network segmentation and controlled communication between application layers.
- Provisioned **Azure Virtual Machines** and **Amazon EC2** instances, integrating **Azure Load Balancer**, **Azure Application Gateway**, and **AWS Application Load Balancer (ALB)** to provide Layer-4/Layer-7 traffic distribution, SSL termination, and high availability.
- Implemented **Azure Traffic Manager** and **AWS Route 53** for DNS-based traffic routing, health checks, and failover while following cloud networking, security, and disaster recovery best practices.
- ***Technologies:*** *MMicrosoft Azure, Azure Virtual Machines, Virtual Network (VNet), Network Security Groups (NSG), Azure Load Balancer, Azure Application Gateway, Azure Traffic Manager, Amazon Web Services (AWS), Amazon EC2, Amazon VPC, Security Groups, Network ACLs, Application Load Balancer (ALB), Amazon Route 53*

Travel Eazy — Production Deployment on Microsoft Azure

- Deployed a production-ready Django-based AI travel recommendation application on an Ubuntu VM using Nginx, Gunicorn, and SSL/HTTPS for secure delivery.
- Configured Azure VNet, NSGs, and Linux server components to ensure secure, reliable, scalable hosting.
- Built a CI/CD pipeline with GitHub Actions that automated build and deployment steps, replacing a fully manual release process.
- Enabled Azure Monitor and Log Analytics for centralized monitoring, alerting, and proactive troubleshooting.
- Configured Nginx as a reverse proxy in front of Gunicorn to handle static file servingand improve request handling performance under load.
- Managed environment-based configuration and application secrets to keep sensitive credentials secure and out of source control across deployments.

Technologies: *Microsoft Azure, Azure VMs, Ubuntu, Python, Django, Nginx, Gunicorn, GitHub Actions, VNet, NSG, Azure Monitor, SSL/HTTPS*

Travel Eazy — Personalized AI-Powered Tourism Platform

- Built an AI-powered travel recommendation web app in Python/Flask, generating personalized suggestions based on user preferences, interests, and health conditions.
- Trained and compared 3 machine learning models (Random Forest, KNN, Gradient Boosting) toselect the best-performing approach for recommendation accuracy.
- Designed the underlying data model to capture user profiles, preferences, and recommendation history to support personalization logic.
- Built a responsive Admin module for managing destinations and content, and a User module for search, browsing, and viewing personalized recommendations.
- Used HTML, CSS, and Bootstrap to deliver an intuitive, accessible interface across both modules.

Technologies: *Python, Flask, Machine Learning, Random Forest, KNN, Gradient Boosting, HTML, CSS, Bootstrap*

EDUCATION

- Bachelor of Computer Applications (BCA) — Mohan Babu University, Tirupati | CGPA: 8.5 | 2025
- Intermediate (BIPC) — Sri Chaitanya Junior College, Kurnool | 79.5% | 2020
- SSC (10th Standard) — Sri Chaitanya E.M. School, Nandyala | CGPA: 8.8 | 2018

CORE STRENGTHS

Strategic Communication • Organizing • Leadership • Teamwork • Adaptive Learning